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CE F231 Fluid Mechanics

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Lecture 21: Fluid Dynamics

Learning Outcome

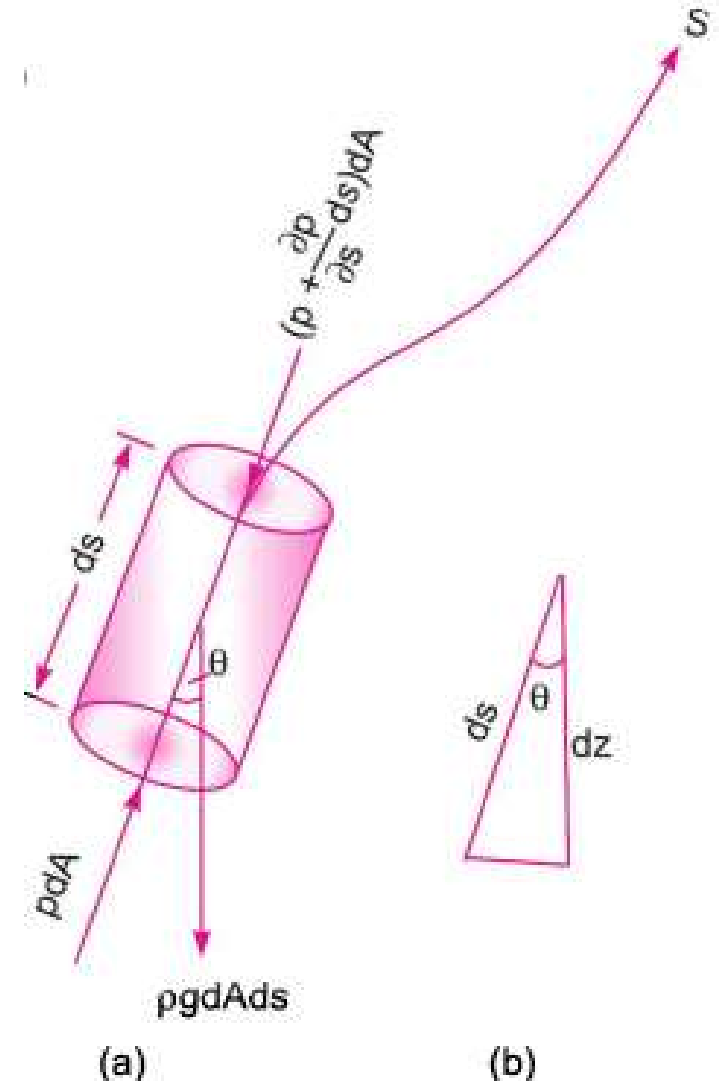
- Euler's equation of motion
- Bernoulli's equation for ideal fluids and real fluids
- Practical applications of Bernoulli's equation
 - Venturi meter, orifice meter and Pitot tube
- Navier Stokes equation
- Momentum equation – force exerted on pipe bend
- Moment of momentum equation
- Vortex Flow

Euler's Equation

$$\frac{dp}{\rho} + g dz + v dv = 0$$

Assumptions:

- Inviscid
- Steady flow
- Incompressible

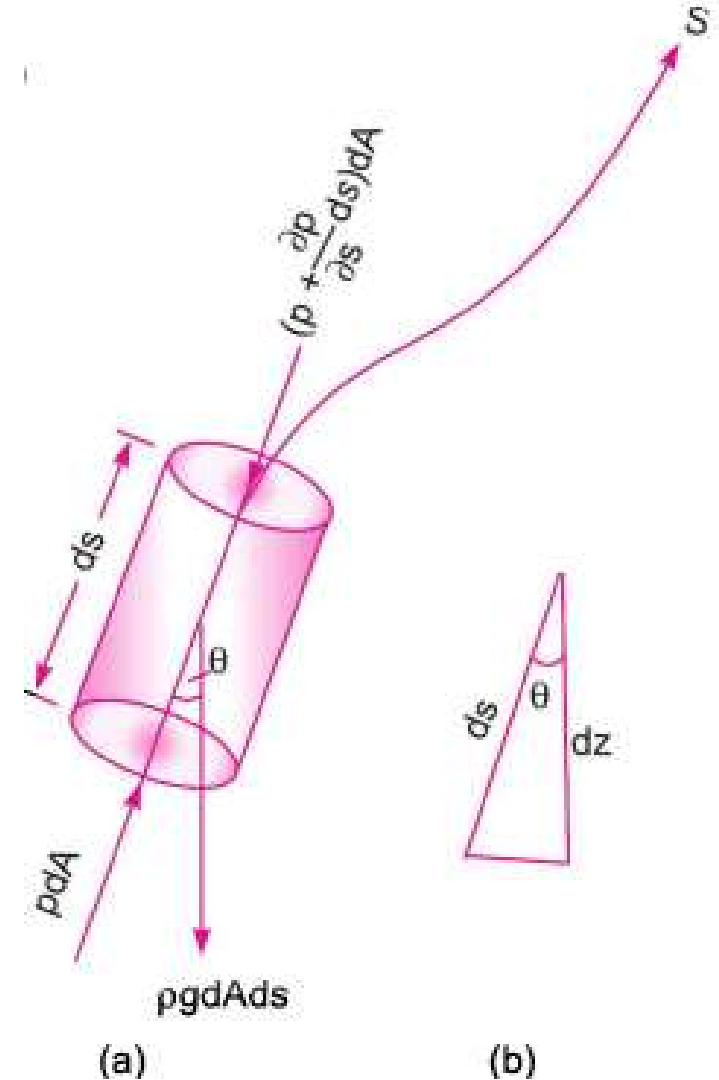


Bernoulli's Equation

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$$

Assumptions:

- Inviscid
- Steady flow
- Incompressible
- Irrotational



Bernoulli's Equation

Bernoulli's equation for real fluids

Direction of flow – section 1 to section 2

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$